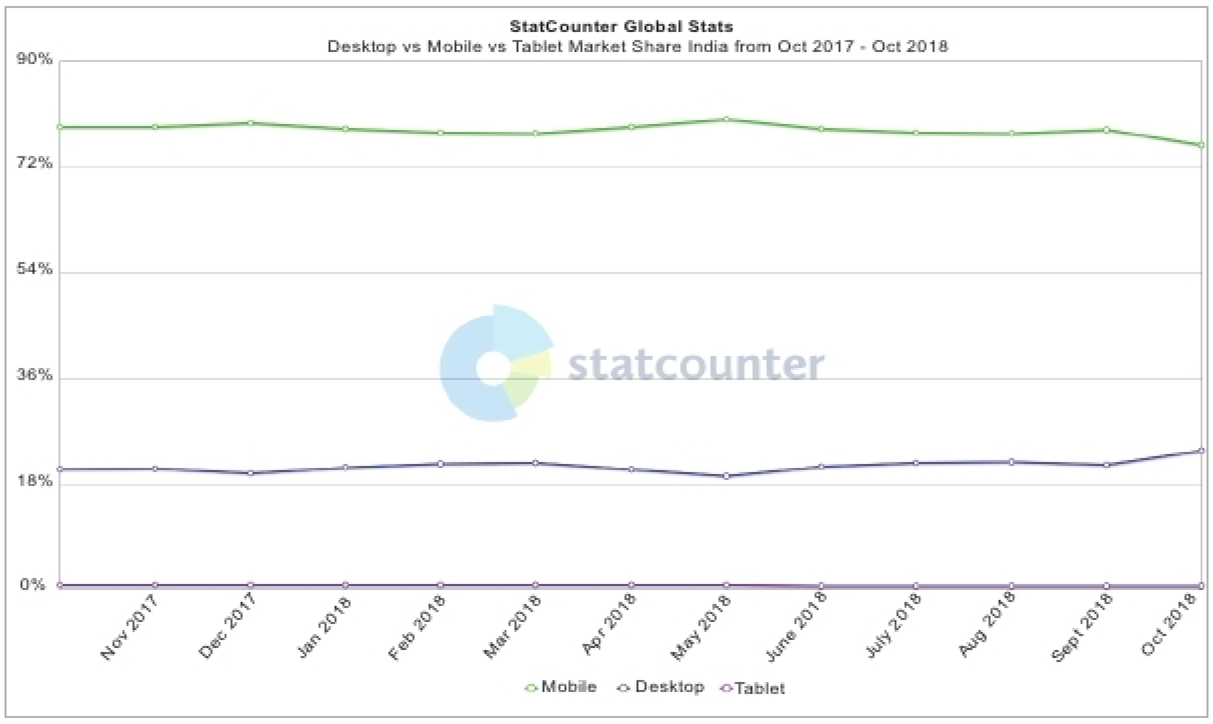

Figure 1: Internet-accessing devices in India

Mobile application development

During and immediately after the floods, many volunteer developers started creating mobile applications to help those affected by the flood. There are many different roles for such apps to play in a post-disaster environment. A very useful function provided by many of these apps was to create a common platform for people affected by the floods and rescue workers to interact with each other. One such app is AmritaKripa, developed by Amrita Vishwa Vidyapeetham (a research and engineering institute). It is an Android app for disaster management that connects flood affected people with relief and rescue service providers. The app is multilingual with support in Malayalam and English.

Many other apps were also proposed immediately after the floods. One project was regarding the documents and certificates lost during the floods. An app was proposed so that people can request for their lost documents through it. In a state like Kerala, where digital penetration is very high, the idea of using mobile apps for disaster management has been highly successful. Figure 1, obtained from the StatCounter Global Stats website, shows the Internet-accessing devices in India (the category 'Desktop' includes laptop computers also). So, in a country like ours, where far more people use mobile phones than desktops, laptops and tablets to access the Internet, mobile apps have a far more important role to play than websites.

Figure 2: Logo of Kotlin

Having discussed the role of mobile apps during and after natural calamities, the important question to answer is, "What mobile application development platform should be used during such times?" Well, I am not going to make any strong recommendations in this category but Kotlin could be a strong contender as a suitable programming language for rapid mobile app development and deployment. Though a relatively new language, for which development started only as recently as 2011, Kotlin (Figure 2) has already proved to be a mature language and environment for developing Android applications quickly. It is currently ranked 41 in the TIOBE index of programming languages and has been steadily gaining popularity over the past years. The speed of apps developed with Kotlin is comparable with those developed with Java, and in fact, it is easier to develop apps using the former. These features make Kotlin a useful app development tool during crisis situations, where fast development and deployment of apps is critical.

Inventory management software

During the floods, a lot of people were stranded in isolated locations and relief camps, needing food and other essential supplies, while at the same time, a lot of food and supplies were ready and in stock, awaiting delivery. Using social media platforms to coordinate inventory management could potentially cause a lot of problems due to the distributed nature of the medium. It is often possible that the same group of people could have made multiple requests, or the same supplies could have been promised to multiple groups by mistake. This can lead to wastage of food and other basic supplies.

Manual inventory management is a challenging issue due to the large scale evacuations during natural calamities. But this problem can be solved easily by the proper deployment of good inventory management software. One open source inventory management application is Apache OFBiz. It is a developer friendly suite of business applications that can be used to handle different scenarios. The common architecture of Apache OFBiz allows developers to easily extend and enhance it to create custom features necessary for inventory management during and after natural disasters. Other such open source apps include

Figure 3: Logo of Apache OFBiz